Name:	Index No.:	CT Group: 10

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CHEMISTRY **9647/03**

Paper 3 Free Response **12 September 2011**

2 hours

Additional Materials: Answer Paper
Data Booklet
Cover Page

READ THESE INSTRUCTONS FIRST

Write your Centre number, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** out of the five questions.

A Data Booklet is provided.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
At then end of the examination, fasten all your work securely together.

Answer any **four** questions

1 **(a)** Solid aluminium chloride sublimes at 180°C. In the vapour phase, an equilibrium is established between aluminium chloride and its dimer.

$$Al_2Cl_6(g) \rightleftharpoons 2AlCl_3(g)$$

At 400ºC and 1 atm, 0.84 g of the equilibrium gaseous mixture occupies 268 cm^3.

(i) Calculate the average relative molecular mass of the gaseous mixture, giving your answer to 1 decimal place.

(ii) Determine the degree of dissociation of Al_2Cl_6, α, using the formula:

$$\alpha = \left(\frac{M_r(Al_2Cl_6) - \text{average } M_r}{\text{average } M_r}\right)$$

(iii) Hence, calculate the value of the equilibrium constant, K_p, for the dissociation of Al_2Cl_6 at 400ºC.

(iv) State and explain the effect of increasing the temperature on the average M_r of the equilibrium mixture.

[7]

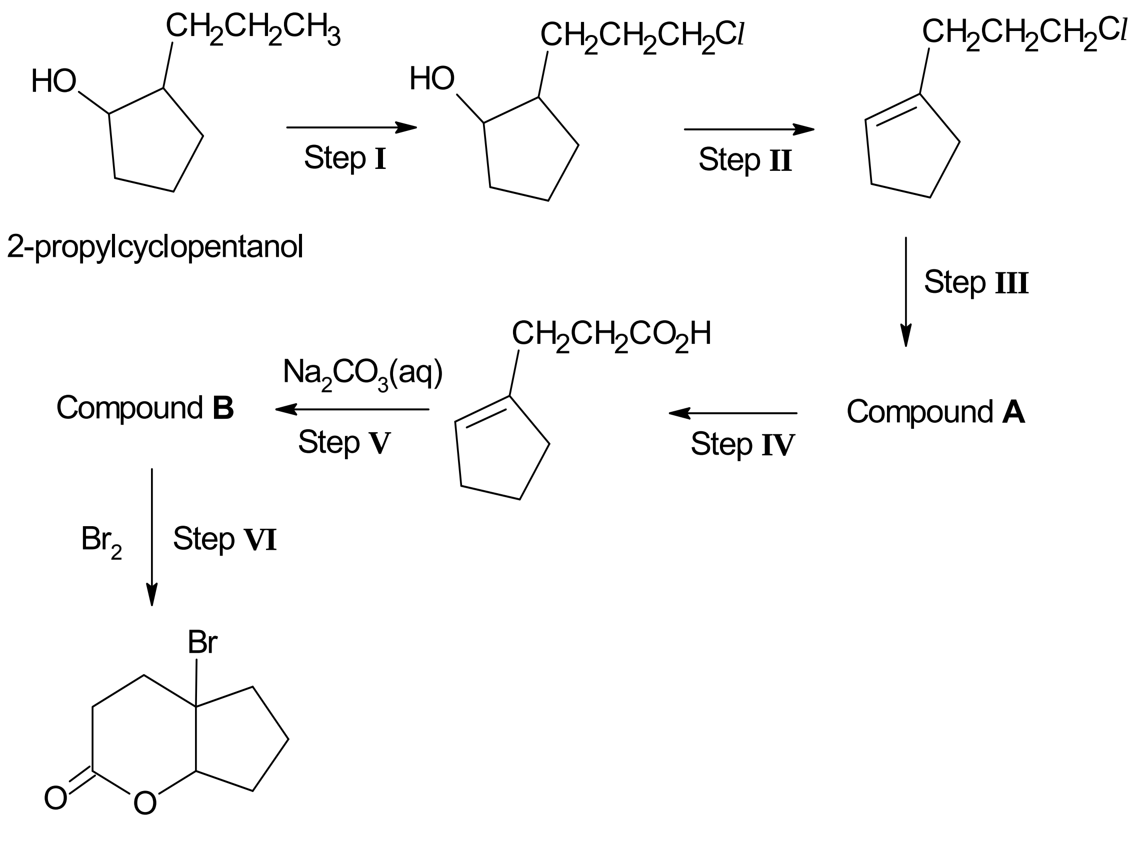
(b) **(i)** Explain what is meant by the term *bond energy*.

(ii) By means of an energy cycle, calculate the average bond energy of the Al–Cl bond using the following data and information from the *Data Booklet*.

Standard enthalpy change of formation of $AlCl_3(g)$ = -585 kJ mol^{-1}
Standard enthalpy change of atomisation of Al = +314 kJ mol^{-1}

[4]

(c) One of the uses of halogens is in the preparation of halogen-containing organic compounds. One particular synthesis is given below.



(i) In step **I**, a sample of 2-propylcyclopentanol and limited amount of chlorine are mixed together and irradiated with ultraviolet light.

The mechanism involves two propagation steps.

(RH represents 2-propylcyclopentanol)

Propagation step 1: $RH + Cl\bullet \rightarrow R\bullet + HCl$
Propagation step 2: $R\bullet + Cl_2 \rightarrow RCl + Cl\bullet$

The respective enthalpy changes of reaction and activation energies for the two propagation steps are given below:

Step	**ΔH / kJ mol^{-1}**	**E_a / kJ mol^{-1}**
Propagation step 1	+4	17
Propagation step 2	-109	4

Using the information provided, sketch the <u>combined</u> energy profile diagram for the two propagation steps of the chlorination of 2-propylcyclopentanol, showing all relevant energy changes and the reaction intermediates.

(ii) Give the structural formulae of compounds **A** and **B**.

(iii) State the reagents and conditions that are required to carry out the reactions in steps **II**, **III** and **IV**.

(iv) Given that step **VI** involves electrophilic addition, propose a suitable mechanism for the reaction.

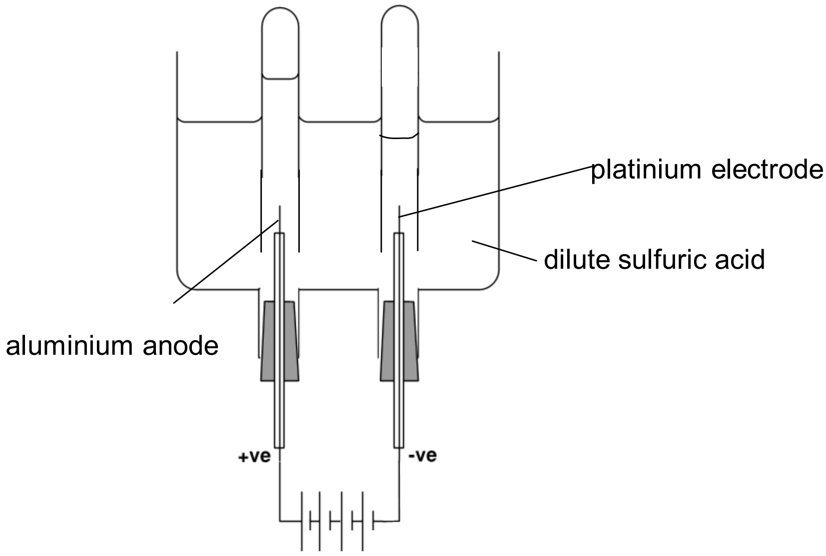
[9]

[Total: 20]

2 **(a)** The anodised aluminium layer is grown by passing a direct current through dilute sulfuric acid, with the aluminium object serving as the anode.

When current is passed through the circuit, the oxygen liberated at the anode will react with aluminium to create a build-up of aluminium oxide.

3.0 A of current is passed through the circuit and the volume of the oxygen collected at the anode after 10 minutes is 40 cm^3.



(i) Give the ion-electron equations to account for the liberation of gas at each electrode.

(ii) Write an equation to show the formation of aluminium oxide.

(iii) Calculate the volume of oxygen that would be produced at room temperature and pressure when an inert electrode is used as the anode.

(iv) Using your answer in **(a)(iii)**, calculate the mass of aluminium oxide produced at the anode.

[7]

(b) **(i)** Aluminum chloride is very soluble in water and its solution has a pH of approximately 3. With the aid of equations, account for the observations.

(ii) What will be observed when aqueous sodium hydroxide is added gradually to aqueous aluminum chloride until it is in excess? Give the formulae of the aluminium-containing species formed in this reaction.

[4]

(c) A recently discovered organic compound **L** possesses a pleasant floral scent and is therefore being used as a component of many essential oils and perfumes. A molecule of **L** has the molecular formula, $C_{10}H_{18}O$.

A sample of **L** has the ability to rotate plane-polarised light.

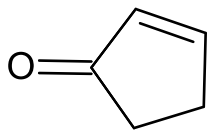
No reaction occurs between **L** and 2,4-dinitrophenylhydrazine. However, **L** produces **M**, $C_{10}H_{22}O$, with hydrogen with nickel catalyst.

No change is observed when **L** is heated with acidified potassium dichromate(VI).

Heating **L** with aluminium oxide produces a mixture of two products. **N**, with molecular formula $C_{10}H_{16}$, is present as the major product.

N reacts with hot concentrated acidified potassium manganate(VII) to produce propanone and **P**, $C_5H_6O_5$. A colourless gas, which forms a white precipitate in limewater, is liberated.

P can also be produced by the reaction of the following compound with hot concentrated acidified potassium manganate(VII).



Deduce the structures of compounds **L**, **M**, **N**, and **P**. Explain the chemistry of the reactions described.

[9]

[Total: 20]

3 **(a)** Human blood is slightly basic, with a normal pH of approximately 7.4. Its pH must be maintained for the effective transport of oxygen.

When CO_2 is dissolved in the blood plasma, a series of equilibria is established.

I $CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$

II $H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$

Equilibrium **II** has a K_a value of 7.90×10^{-7} mol dm^{-3} and is responsible for buffering the pH of the blood plasma at 7.4.

In the lungs, excess dissolved $CO_2(aq)$ is exhaled as $CO_2(g)$ according to the following equilibrium.

III $CO_2(aq) \rightleftharpoons CO_2(g)$

During heavy exercise, lactic acid is released into the blood and is buffered by the blood plasma. The buffering action of the blood plasma will eventually lead to an increased breathing rate.

(i) State what is meant by a *buffer solution*

(ii) Use the data above to determine the $[HCO_3^-(aq)]/[H_2CO_3(aq)]$ ratio in the blood plasma.

(iii) Write an equation to show how lactic acid is removed from the body. Use RCO_2H to represent lactic acid.

(iv) Hence, by considering the effect on equilibria **I** and **III**, explain why the process of removing lactic acid causes the breathing rate to increase.

(v) A student wishes to simulate the blood plasma buffer by mixing 25.0 cm^3 of 0.20 mol dm^{-3} H_2CO_3 solution with 0.30 mol dm^{-3} NaOH solution. The neutralisation that takes place is represented by the following equation.

$$NaOH + H_2CO_3 \rightarrow NaHCO_3 + H_2O$$

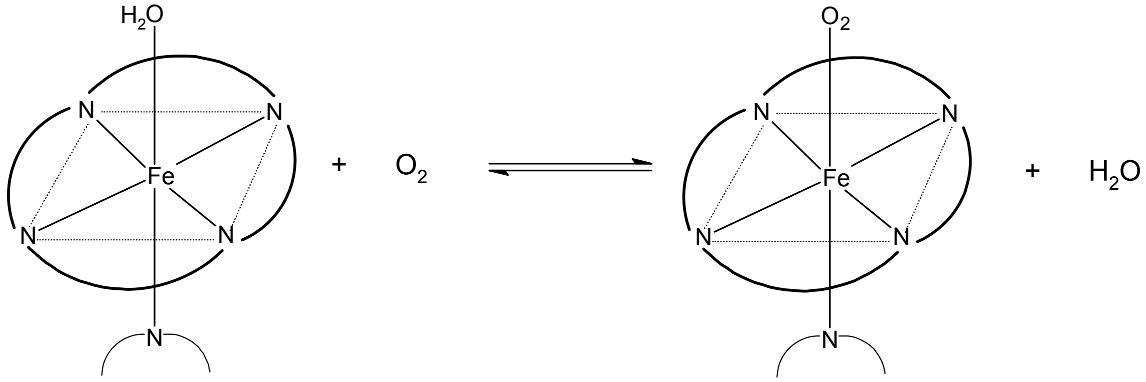
Using your answer in **(a)(ii)**, calculate the volume of NaOH that should be added.

[8]

(b) Haemoglobin (Hb) is an oxygen carrier protein that circulates around the body within red blood cells.

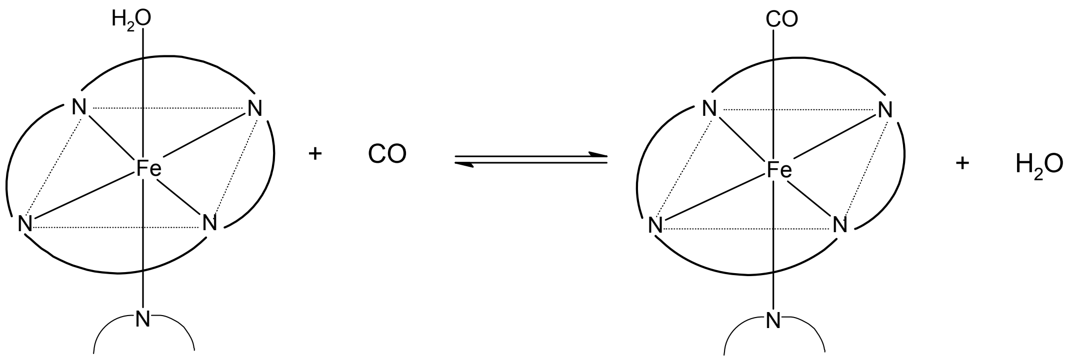
Each Hb molecule contains four subunits ($\alpha_2\beta_2$), which each contains a haem group responsible for binding oxygen.

A haem is an iron chelate of a tetrapyrrole macrocycle. The iron ion is situated in the centre of the planar macrocycle, and is the site of oxygen binding. Haem can be represented schematically as shown.



(i) Describe the bonding around the iron ion.

(ii) In the blood of smokers, another form of Hb, carboxyhaemoglobin, is also common.



CO is found to have 245 times more affinity for haemoglobin than O_2. Explain how CO gives rise to toxicity/ poisoning in the blood of smokers.

(iii) Explain why CO is unable to displace the tetrapyrrole macrocycle from the iron centre.

[4]

(c) The secondary structure of haemoglobin consists of segments of α-helices. Repeated folding of the secondary structure gives rise to a highly globular protein subunit. Four globular protein subunits ($\alpha_2\beta_2$) assemble to form haemoglobin.

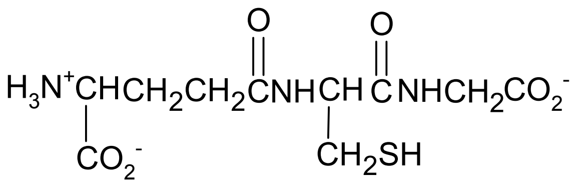
(i) Describe the α–helix structure with the aid of a suitable diagram.

(ii) The amino acids which are on the exterior of haemoglobin are likely to interact strongly with water molecules.

amino acid	**formula of side chain (R in $RCH(NH_2)CO_2H$)**
serine	$—CH_2OH$
aspartic acid	$—CH_2CO_2H$
valine	$—CH(CH_3)_2$
lysine	$— CH_2CH_2CH_2CH_2NH_2$
phenylalanine	$—CH_2C_6H_5$

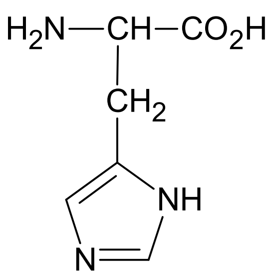
Identify two amino acids from the above table that are likely to have this property. Explain your choice.

(iii) A mutated tripeptide segment of an α–polypeptide chain of haemoglobin was obtained via partial hydrolysis.

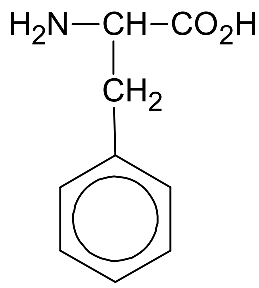


What is unusual about the peptide bond formed by the N–terminal amino acid in this tripeptide?

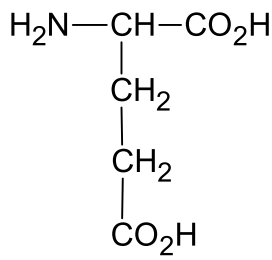
(iv) Another tripeptide, his–phe–glu, from a β–polypeptide chain was hydrolysed. The resulting solution was added to an excess of a buffer solution of **pH 5.5** and placed at the centre of the plate. A potential difference was then applied across the plate.



his



phe

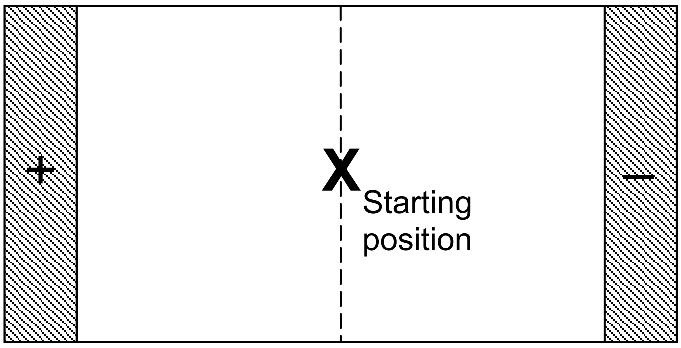


glu

	his	phe	glu
pI	7.6	5.5	3.1

pK_a (2^o amine): 4.0;
pK_a (N of C=N): 6.0

(I) Draw the structural formula of the resulting histidine at this pH.
(II) Copy the diagram shown below onto your writing paper and indicate the relative positions of the three amino acids on the diagram.



[8]

[Total: 20]

4 Silver and lead form a series of halides of general formula AgX and PbX_2. The halides of these two cations are sparingly soluble in water at 25°C.

Salt	Solubility product / K_{sp}
Ag*Cl*	2.00×10^{-10} mol^2 dm^{-6}
Pb*Cl*$_2$	1.70×10^{-5} mol^3 dm^{-9}
$PbBr_2$	6.60×10^{-6} mol^3 dm^{-9}

(a) A solution of 100 cm^3 of 0.120 mol dm^{-3} $AgNO_3$(aq) was accidentally mixed with 50 cm^3 of 0.120 mol dm^{-3} $Pb(NO_3)_2$(aq). Two methods were suggested to separate the cations.

(i) In the first method, 0.180 g of solid Na*Cl* was added to the mixture. Show, by means of calculation, how the two cations can be separated by selective precipitation.

(ii) The alternative method of separating the two metal ions is to add **excess** aqueous Na*Cl* to the mixture in **(a)** followed by excess NH_3(aq) and then filtration.

Construct ionic equations for
(I) the reactions between the mixture in **(a)** and excess Na*Cl*(aq).
(II) the reaction that takes place when excess NH_3(aq) is added.

[7]

(b) When a precipitate is formed, ΔG^{θ}_{ppt}, in J mol^{-1}, is given by the following expression.

$$\Delta G^{\theta}_{ppt} = 2.303\ RT \log K_{sp}$$

(i) Use the data above to calculate ΔG^{θ}_{ppt} for lead(II) bromide.

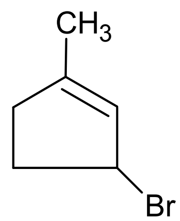
(ii) The standard enthalpy change of precipitation of lead(II) bromide is -279 kJ mol^{-1}. Hence, calculate ΔS^{θ}_{ppt}, in J mol^{-1} K^{-1}, for lead(II) bromide at 298K.

(iii) Explain the significance of the sign of your answer in **(b)(ii)**.

(iv) Assuming ΔS_{ppt} does not change with temperature, state the effect of increasing temperature on ΔG_{ppt} and hence solubility of lead(II) bromide.

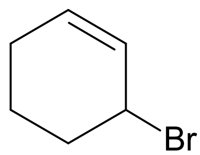
[4]

(c) Student **A** from Pioneer Junior College found a bottle of a liquid substance that is labeled, C_6H_9Br. He suggested that the unknown has the following structure.



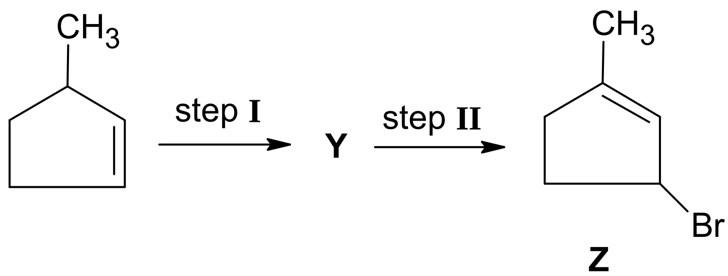
(i) Plan **two** chemical reactions that can positively confirm the suggested structure. Include your observations and deductions for each test.

(ii) Student **B** from the same college thinks otherwise and proposed that the unknown liquid has the structure below.



Propose a chemical test which would enable you to distinguish between them. Include your observations.

(iii) Compound **Z** can be formed by the reaction sequence below.



Suggest the structural formula of compound **Y**. State the reagents and conditions in each of the steps.

[9]

[Total: 20]

5 **(a)** The standard redox potential of manganate(VII) ions is given below:

$$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O \qquad E^\theta = +1.52V$$

(i) What you understand by the term *standard electrode potential*?

(ii) Draw a fully labelled sketch of how you would determine the standard electrode potential of the $MnO_4^-(aq) \mid Mn^{2+}(aq)$ electrode system.

[3]

(b) Potassium manganate(VII) is also frequently used in titrimetric analysis due to its strong oxidising property.

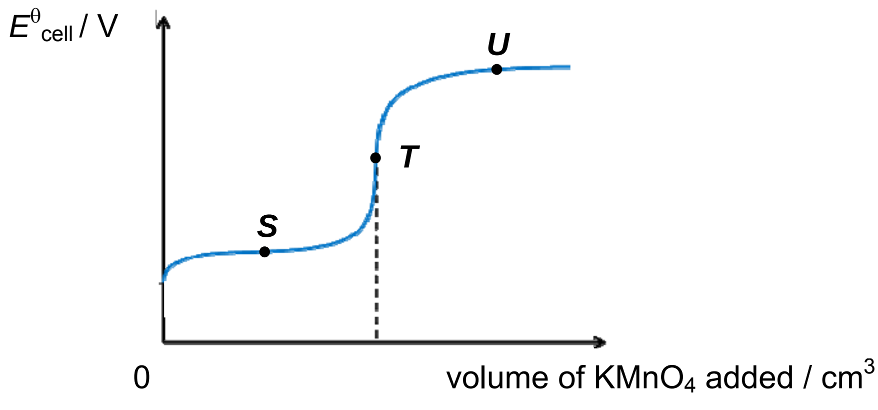
A student was tasked to determine the amount of iron(II) salts present in a nutritional supplement pill. He decided to carry out a titration with a standard solution of acidified potassium manganate(VII) that he prepared.

(i) Write a balanced equation for the reaction of MnO_4^- and Fe^{2+} in acid solution.

(ii) The acid that was used to prepare the standard solution was sulfuric acid. Using appropriate data from the *Data Booklet*, explain why hydrochloric acid could **not** be used.

(iii) Suggest why potassium manganate(VII) rather than potassium dichromate(VI) was used in the titration.

One such nutritional supplement pill containing Fe^{2+} was dissolved in a volume of dilute sulfuric acid and titrated against 0.0200 mol dm^{-3} $KMnO_4$. The E^θ_{cell} was measured against a standard hydrogen electrode and the following graph was obtained.



(iv) Using suitable data listed in the *Data Booklet*, suggest approximate E^θ values, in volts, for the points ***S*** and ***U*** on the graph.

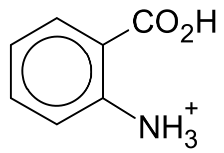
(v) State an important assumption that was made to estimate the answers in part **(b)(iv)**.

(vi) State the colour of the solution at point ***T***.

[8]

(c) Compound **H** has the molecular formula, C_9H_9NO.

- **H** is insoluble in aqueous sodium hydroxide and hydrochloric acid at room temperature.
- When **H** is heated with acidified potassium manganate(VII), a compound with the following structure is obtained.



Compound **H** can be prepared by a series of reactions starting from optically active compound **J**, $C_9H_{12}O$.

- Compound **J**, $C_9H_{12}O$, reacts with concentrated nitric acid and sulfuric acid to give **K**, $C_9H_{11}NO_3$.
- Treating compound **K** with hot acidified potassium dichromate(VI) gives compound **L**, $C_9H_9NO_4$.
- Compound **L** then reacts with phosphorus pentachloride to give **M**, $C_9H_8NO_3Cl$.
- Reduction of **M** under suitable conditions yields **N**, $C_9H_{10}NOCl$, which undergoes immediate reaction by eliminating a molecule of hydrogen chloride to give **H**.

Deduce the structures of **H**, **J**, **K**, **L**, **M** and **N**. Explain the chemistry of the reactions described.

[9]

[Total: 20]

End of Paper